Chan 2021 SCLC — analysis summary

GO / NO-GO GATES

Gate 1 — batch mixing & continuity → GO

24/34 samples pass the mult-gap threshold (≥37 cells); 50,886 malignant cells share one continuous, isotropic manifold (UMAP aspect 1.04). Samples & SCLC-A/N/P subtypes intermix (θ sweep 0.83→0.22); cell-cycle confound regressed out (mean |corr| 0.32→0.01).

Gate 2 — embryonic module geography → GO (non-branching)

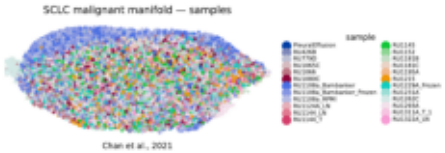
Single monotone axis: Gast & pSom co-vary (r=+0.73) and decline from the root (ρ −0.24/−0.45); Crest opposes both (r=−0.71/−0.88) and rises to a terminal readout (ρ +0.24). No bifurcation.

Gate 3 — hub cross-repression → NA

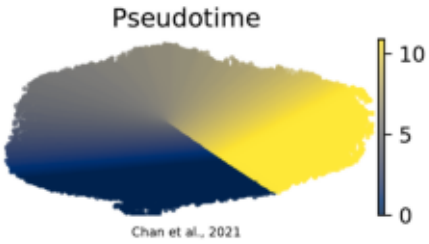
Non-branching manifold has no bifurcation/hub, so the cross-repression test does not apply.

STEP FIGURES

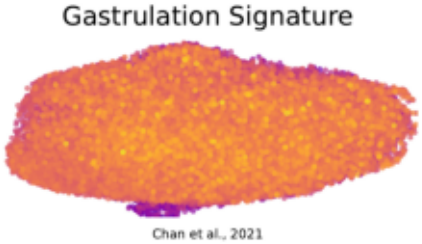
Gate 1: integration



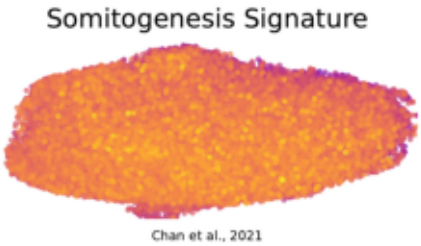
Trajectory / root



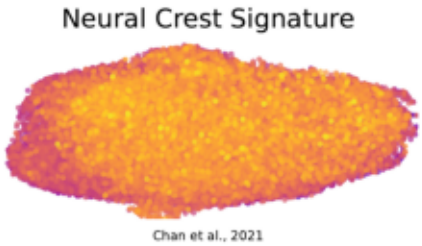
Gast



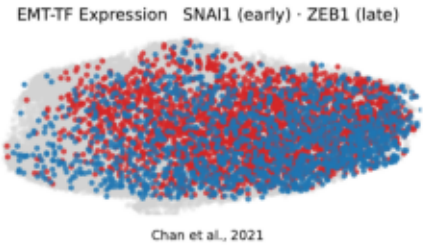
pSom



Crest



EMT-TFs



Dataset. Small-cell lung cancer (Chan et al., 2021), 10x. Master scope 37 patients / 86,662 cells; malignant compartment 51,052 cells (SCLC-A/N/P + minor NSCLC); 50,886 across 24 samples enter the trajectory.

Verdict. ✓ Pattern reproduces as the non-branching case (G1 GO / G2 GO / G3 NA): a single monotone gastrulation/somitogenesis → terminal neural-crest gradient, not resolved into Puram's two competing arms. SCLC is intensely proliferative, so a raw-embedding curve tracks cell cycle (ρ −0.79); regressing the cell-cycle scores out (post ρ=0.00) recovers developmental order. EMT SNAI1 → ZEB1.

Parameter choices

Min cells/sample	≥37 (mult-gap 92→15; keeps 24/34)	**Integration**	Harmony θ=5; PCA50→20; spread=1.3
Cell-cycle	G1S+G2M regressed out (0.32→0.01)	**Trajectory**	scFates curve N=30 λ=0.3 μ=100 (SA 2.06)
Root node	tip 0 = Gast/pSom-high, Crest-low	**EMT-TFs**	SNAI1 (root) → ZEB1 (terminus)